

Plausible Outliers and Implausible Inliers

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Secondary data analyses of large administrative data sets can provide important information that complements the information provided by trials and prospective observational studies. There is a wide range of expertise and experience in researchers involved these studies, and it can be easy to underestimate the challenges of working with these data.

In this issue, Annis et al. (1) report a systematic review of body weight measures and other aspects of methods in analyses of weight, using Veterans Health Affairs administrative data sets. The authors found substantial variation in methods and reporting in these studies and noted the potential for improving the quality of analyses through standardization. This discussion will focus on one aspect of methods for these data sets: how to handle outliers and implausible values.

One potential area of standardization is terminology. This author believes it is important to use the terms outliers and implausible values to refer to related but distinct concepts (2). Outliers are values that are substantially different from most values in a data set, a reference population, or longitudinal data from a single person. Implausible values are impossible or extremely unlikely to occur given our understanding of the human body in the population being studied.

Using this terminology, an outlier may or may not be implausible; a height of 6'10" in a man in the general population is a plausible outlier. Similarly, an implausible value may be an outlier or an inlier (i.e., not an outlier) (2,3); the third measurement in the following series of four heights for the same woman would be an inlier compared with a reference population but is still implausible: 70", 70.5", 65", and 70". Finally, if the same woman had weights recorded of 110 kg, 109 kg, 110 kg, and 67 kg, the final weight would be a longitudinal outlier for the individual and a reference population inlier, and it may or may not be plausible. Longitudinal data, when available, provide important information. In this example, the timing of the four measurements or any additional measurements (e.g., a fifth weight of 68 kg) would help to determine plausibility.

Values that are identified as implausible should not be included in analyses; by definition, these values are not thought to be representative of a study participant's true measurement. Outliers, however, should not automatically be excluded. Careful efforts to distinguish implausible values from plausible outliers are particularly important when patients with more extreme values, such as severe obesity, failure to thrive, or rapid weight gain or loss, are a focus of the study. It has been shown that the exclusion of children with outlying BMIs (a modified BMI z score

of +5 to +8), which are now thought to be plausible, can bias estimates of the prevalence of severe obesity (4).

Currently, guidance for handling outliers and implausible values in adults would need to be limited primarily to 1) a discussion of relevant definitions and concepts and 2) recommendations to report the methods and rationale for how these values are identified and treated. There is little evidence to support any standardized guidance for identifying implausible values in adults. More research is available for pediatric data, and the advantages and limitations of existing methods for children could be presented in guidelines while further advancement of these methods continues (2,4-9). Any guidelines regarding plausible outliers should reflect that the treatment of these values depends on the research question. Recommendations for reporting sensitivity analyses with varied identification and treatment of implausible and outlying values could improve interpretation of individual studies and build the evidence base for future guidelines.

Finally, no amount of data preparation can turn low-quality measurements into high-quality measurements. Guidance that includes recommendations for higher quality measurements that are feasible in a clinical setting could improve both clinical care and research. 

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